

**AMRITA VISHWA VIDYAPEETHAM,
AMRITAPURI**

**B. Tech CSE (ARTIFICIAL
INTELLIGENCE)**

LITERATURE REVIEW SUBMISSION

Group No: 5

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1. Problem Definition

The research problem focuses on leveraging Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques to extract, analyze, and represent structured knowledge from Ayurvedic texts written in Sanskrit. Classical texts like the Charaka Saṃhitā and Suśruta Saṃhitā are rich sources of medical knowledge but pose challenges due to complex linguistic constructs such as sandhi, samāsa, free word order, and the morphological richness of Sanskrit. The goal is to design an AI-based framework that performs both syntactic and semantic analysis of these texts, leading to the construction of a queryable Knowledge Graph (KG). This KG will include entities such as DRAVYA (substances/herbs), ROGA (diseases), LAKSHANA (symptoms), and DOSHA (bio-energetic principles), along with their inter-relationships. The ultimate objective is to bridge traditional Ayurvedic wisdom with modern computational tools for knowledge retrieval, decision support, and further research in integrative medicine.

2. Literature Review Summary Table

No.	Author(s) & Proceedings	Title	Year	Dataset Used	Methodology / Approach	Limitations & Future Work
1	Sandhan et al., ACL Demo 2023	SanskritShala: A Neural Toolkit and Web Interface for Sanskrit NLP	2023	Various Sanskrit corpora	Neural NLP toolkit with web interface	Needs Ayurveda-specific extensions
2	Terdalkar et al., arXiv/WSC 2022	Ontology-Driven Knowledge Graphs for Ayurvedic Texts	2022	Ayurvedic texts	Ontology-driven KG construction	Limited automation, manual effort
3	Terdalkar et al., WSC Poster 2023	Āyurjñānam: Queryable Knowledge Graph for Ayurvedic Texts	2023	Annotated KG	Queryable KG demo	Requires large-scale expansion
4	Nedungadi et al., JAIM 2023	Mapping Research Trends in Ayurveda: A Bibliometric Study	2023	Bibliometric databases	Bibliometric analysis	Lacks computational methods
5	Krishnan,	Digital	2023	Digital	Segmentation	Not

	Kulkarni, Huet, WSC 2023	Corpus of Sanskrit: Towards Improved Segmentation and Normalization		Corpus of Sanskrit (DCS)	& normalization techniques	Ayurveda-specific
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3. Review of Selected Papers

Sandhan et al. (2023) presented SanskritShala, a neural NLP toolkit with a web interface. It integrates several pre-trained models for Sanskrit processing and makes them accessible to researchers. Although powerful, it is not tailored specifically to Ayurveda.

Terdalkar et al. (2022) proposed ontology-driven knowledge graphs for Ayurvedic texts. Their approach emphasizes the semantic structure of Ayurveda but relies heavily on manual work, highlighting the need for automated methods.

Terdalkar et al. (2023) introduced Āyurjñānam, a queryable knowledge graph for Ayurvedic texts. It provides a useful demonstration of knowledge graph querying but remains limited in scope and needs scaling to larger corpora.

Nedungadi et al. (2023) performed a bibliometric study to map Ayurveda research trends from 1993–2022. The study provides valuable insights into research growth and patterns but does not contribute computational tools or resources.

Krishnan et al. (2023) improved segmentation and normalization in the Digital Corpus of Sanskrit. Their work enhances textual preprocessing for Sanskrit but is not specialized for Ayurvedic corpora, leaving gaps for adaptation.

4. Identified Research Gaps

- Lack of annotated Ayurvedic corpora
- Limited automation in knowledge graph construction
- Existing NLP tools not Ayurveda-specific
- Bibliometric studies lack computational contribution
- Normalization/segmentation not adapted to Ayurveda

5. Novelty / Contribution of Proposed Work

- Creation of annotated Ayurvedic corpus
- Fine-tuning Sanskrit PLMs (ByT5-Sanskrit) for Ayurveda
- Semi-automatic KG construction
- Neo4j-based queryable KG
- Open-source release for reproducibility

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Sign: _____